

# Solar Fe II appendix

Sun · Fe II · downloadable appendix

Fe.json v1.221 · 30 Fe II products · 160 total

Solar Fe I anchor:  $7.477 \pm 0.052$

HARPS · Reference Grade · Amarsi 3D-NLTE ·  $n = 21$ . The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

## Highlighted band products

### near-UV

**$7.402 \pm 0.121$**

Kitt Peak · Synth · 1D-LTE · Deep Grade ·  $n = 12$

Systematic under development — the abundance is current (post-RYA-1207 molecular opacity) but the uncertainty budget is not closed: component wiring and the blends bound are pending RYA-1226.

### VIS

**$7.531 \pm 0.108$**

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade ·  $n = 2$

## Error-bar forest

**How to read this forest.** Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Solid bars show statistical uncertainty from the line measurements; wireframes show systematic uncertainty from the product’s adopted error budget, which does not average away with more lines. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool’s provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Unmeasured model rows remain N/A.

Grades and model labels

**Reference Grade.** An externally published reference line pool, using that reference’s line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a promise that every line has an individual primary-laboratory uncertainty.

**Deep Grade.** The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.

**1D-LTE** uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium.

**1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

**Synthesis** fits the spectral line profile. See [Methodology](#) for the full model, line-selection and uncertainty procedures.

Asplund, Amarsi & Grevesse 2021  $7.46 \pm 0.04$  · Lodders+ 2025  $7.49 \pm 0.01$

blue solid = statistical  $\sigma$  · wireframe = systematic  $\sigma$  · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders · N/A = no product for this model

**near-UV — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.26-7.78**

EW · 1D-LTE  
no product

N/A

**near-UV — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected · DEEPGRADED (no graded product in this band)  
A(Fe) dex; axis 7.26-7.78**

|                                                    |                                                                                    |                                  |
|----------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------|
| Synth · 1D-LTE<br>Deep Grade · n=12                |  | 7.617<br>±0.052 stat ±0.110 syst |
| Synth · 1D-LTE · Gerber<br>Deep Grade · n=12       |  | 7.622<br>±0.053 stat ±0.112 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                    | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=7    |  | 7.635<br>±0.081 stat ±0.110 syst |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                    | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                    | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                    | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                    | N/A                              |

**near-UV — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected  
A(Fe) dex; axis 7.26-7.78**

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

**near-UV — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected · DEEPGRADED (no graded product in this band)  
A(Fe) dex; axis 7.26-7.78**

|                                                    |                                                                                   |                                  |
|----------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                   | N/A                              |
| Synth · 1D-LTE<br>Deep Grade · n=12                |  | 7.402<br>±0.047 stat ±0.111 syst |
| Synth · 1D-LTE · Gerber<br>Deep Grade · n=12       |  | 7.422<br>±0.045 stat ±0.118 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                   | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=7    |  | 7.404<br>±0.051 stat ±0.110 syst |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                   | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                   | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                   | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                   | N/A                              |

**near-UV — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected  
A(Fe) dex; axis 7.26-7.78**

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

near-UV axis: 7.26 · 7.39 · 7.52 · 7.65 · 7.78 A(Fe) dex

**VIS — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected ·  
DEEPGRADED (no graded product in this band)  
A(Fe) dex; axis 7.15-8.12**

|                                                    |                                                                                   |                                  |
|----------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                   | N/A                              |
| Synth · 1D-LTE<br>Deep Grade · n=8                 |  | 7.571<br>±0.147 stat ±0.108 syst |
| Synth · 1D-LTE · Gerber<br>Deep Grade · n=8        |  | 7.567<br>±0.151 stat ±0.105 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                   | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=2    |  | 7.324<br>±0.139 stat ±0.065 syst |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                   | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                   | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                   | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                   | N/A                              |

**VIS — Kitt Peak — Kurucz 2005 corrected solar\_kpno\_kurucz2005\_corrected  
A(Fe) dex; axis 7.15-8.12**

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

**VIS — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected · DEEPGRADED**  
**(no graded product in this band)**  
**A(Fe) dex; axis 7.15-8.12**

|                                                    |                                                                                   |                                  |
|----------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                   | N/A                              |
| Synth · 1D-LTE<br>Deep Grade · n=8                 |  | 7.560<br>±0.140 stat ±0.101 syst |
| Synth · 1D-LTE · Gerber<br>Deep Grade · n=8        |  | 7.556<br>±0.144 stat ±0.106 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                   | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=2    |  | 7.328<br>±0.131 stat ±0.065 syst |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                   | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                   | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                   | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                   | N/A                              |

**VIS — Kitt Peak — 1984 Molecfit corrected solar\_kpno\_molecfit\_corrected**  
**A(Fe) dex; axis 7.15-8.12**

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

## VIS — HARPS solar\_harps\_molecfits\_corrected · DEEPGRADED (no graded product in this band)

### A(Fe) dex; axis 7.15-8.12

|                                                    |                                                                                    |                                  |
|----------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------|
| EW · 1D-LTE<br>no product                          |                                                                                    | N/A                              |
| Synth · 1D-LTE<br>Deep Grade · n=8                 |  | 7.909<br>±0.164 stat ±0.091 syst |
| Synth · 1D-LTE · Gerber<br>Deep Grade · n=8        |  | 7.925<br>±0.167 stat ±0.101 syst |
| EW · 1D-NLTE · Bergemann<br>no product             |                                                                                    | N/A                              |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=2    |  | 7.531<br>±0.086 stat ±0.065 syst |
| Synth · 1D-NLTE · Gerber<br>no product             |                                                                                    | N/A                              |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |                                                                                    | N/A                              |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |                                                                                    | N/A                              |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |                                                                                    | N/A                              |

## VIS — HARPS solar\_harps\_molecfits\_corrected

### A(Fe) dex; axis 7.15-8.12

|                                                    |  |     |
|----------------------------------------------------|--|-----|
| EW · 1D-LTE<br>no product                          |  | N/A |
| Synth · 1D-LTE<br>no product                       |  | N/A |
| Synth · 1D-LTE · Gerber<br>no product              |  | N/A |
| EW · 1D-NLTE · Bergemann<br>no product             |  | N/A |
| Synth · 1D-NLTE · Bergemann<br>no product          |  | N/A |
| Synth · 1D-NLTE · Gerber<br>no product             |  | N/A |
| Synth · <3D>-LTE · Gerber · stagger<br>no product  |  | N/A |
| Synth · <3D>-NLTE · Gerber · stagger<br>no product |  | N/A |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>no product    |  | N/A |

VIS axis: 7.15 · 7.40 · 7.64 · 7.88 · 8.12 A(Fe) dex

Read all product values, including alternate line sets and experiments

HARPS · Synth · 1D-LTE · Deep Grade · solar\_harps\_molecfi\_t\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.909 \pm 0.187$  reported total;  $n=8$ ;  $\xi$  MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_harps\_molecfi\_t\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.531 \pm 0.108$  reported total;  $n=2$ ;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

HARPS · Synth · 1D-LTE · Gerber · Deep Grade · solar\_harps\_molecfi\_t\_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber:  $7.925 \pm 0.195$  reported total;  $n=8$ ;  $\xi$  MEASURED

HARPS · Synth · 1D-LTE · Reference Grade · solar\_harps\_molecfi\_t\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.909 \pm 0.187$  reported total;  $n=8$ ;  $\xi$  MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_harps\_molecfi\_t\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.531 \pm 0.108$  reported total;  $n=2$ ;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

HARPS · Synth · 1D-LTE · Gerber · Reference Grade · solar\_harps\_molecfi\_t\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.925 \pm 0.195$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.571 \pm 0.183$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.324 \pm 0.153$  reported total;  $n=2$ ;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber:  $7.567 \pm 0.184$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.571 \pm 0.183$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.324 \pm 0.153$  reported total;  $n=2$ ;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.567 \pm 0.184$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar\_kpno\_molecfi\_t\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.560 \pm 0.173$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_kpno\_molecfi\_t\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.328 \pm 0.146$  reported total;  $n=2$ ;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar\_kpno\_molecfi\_t\_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber:  $7.556 \pm 0.179$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_molecfi\_t\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.560 \pm 0.173$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_molecfi\_t\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.328 \pm 0.146$  reported total;  $n=2$ ;  $\xi$  UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_molecfi\_t\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.556 \pm 0.179$  reported total;  $n=8$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.617 \pm 0.122$  reported total;  $n=12$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.635 \pm 0.137$  reported total;  $n=7$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar\_kpno\_kurucz2005\_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber:  $7.622 \pm 0.124$  reported total;  $n=12$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.617 \pm 0.126$  reported total;  $n=12$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.635 \pm 0.140$  reported total;  $n=7$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_kurucz2005\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.621 \pm 0.124$  reported total;  $n=12$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar\_kpno\_molecfit\_corrected · DEEPGRADED · SYNTH · 1D-LTE:  $7.402 \pm 0.121$  reported total;  $n=12$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar\_kpno\_molecfit\_corrected · DEEPGRADED · SYNTH · ENGINE-A:  $7.404 \pm 0.121$  reported total;  $n=7$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar\_kpno\_molecfit\_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber:  $7.422 \pm 0.126$  reported total;  $n=12$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · 1D-LTE:  $7.402 \pm 0.124$  reported total;  $n=12$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · ENGINE-A:  $7.404 \pm 0.125$  reported total;  $n=7$ ;  $\xi$  MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar\_kpno\_molecfit\_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber:  $7.415 \pm 0.126$  reported total;  $n=12$ ;  $\xi$  MEASURED

## Solar iron context

Fe II traces singly ionized iron and is retained as an ionization and consistency diagnostic alongside Fe I, not an independent headline to average with it. The current resolved pool contains strong, saturation-sensitive lines and documented blend exclusions; the diagnostics below distinguish those limitations from model and holding differences.

## Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma\_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline, which rests on the VIS anchor 7.466 (RYA-161: nothing calibrated to a target).

## Frankenstein — experimental, pending the Bride

4 experimental products remain available in the Fe I product values and downloads.

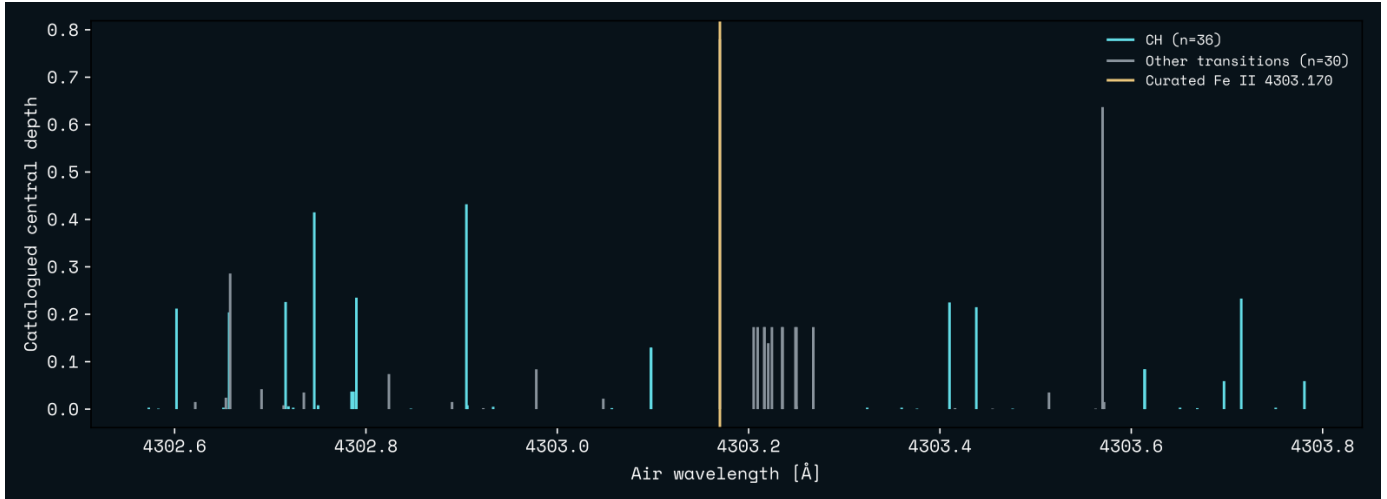
RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

## Saturation characterization

Committed proof census: 120 line/holding records; 100% have observed depth > 0.70. Median observed  $\log_{10}(\text{EW}[\text{m}\text{\AA}]/\lambda[\text{\AA}]) = -1.252$ . In consistent Å units this is -4.252. These diagnostics characterize the audited saturated pool, not every newly added product.

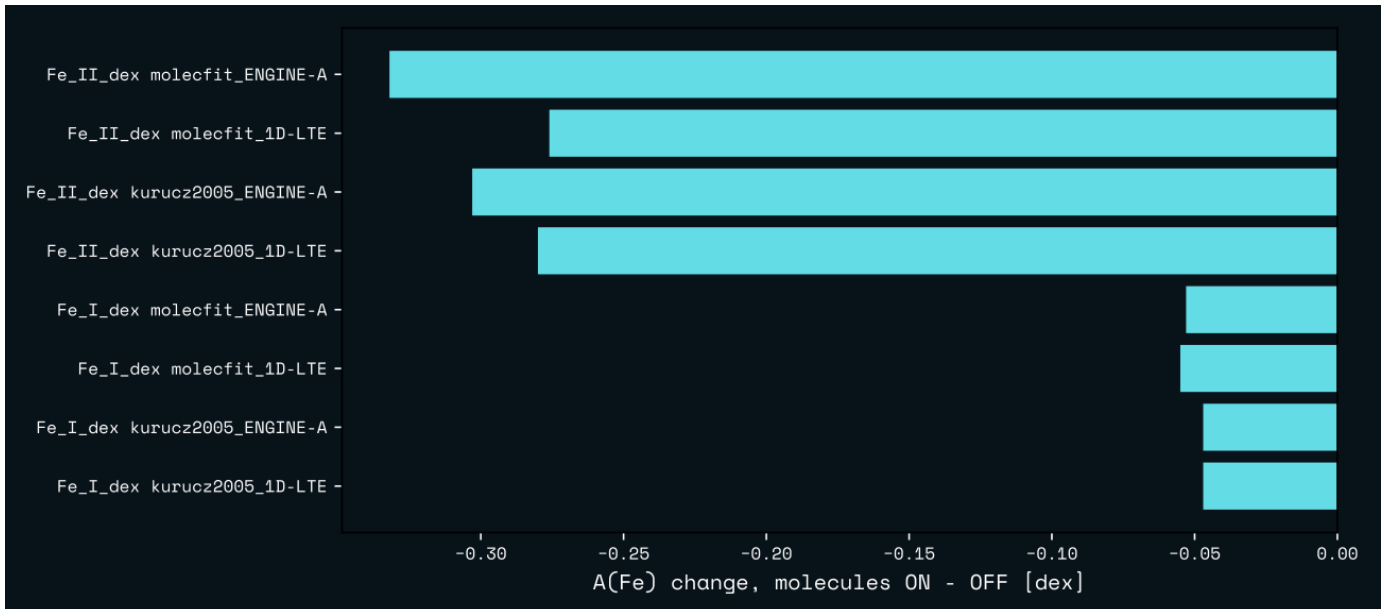
## Problem-line and model diagnostics

### CH G-band artifact



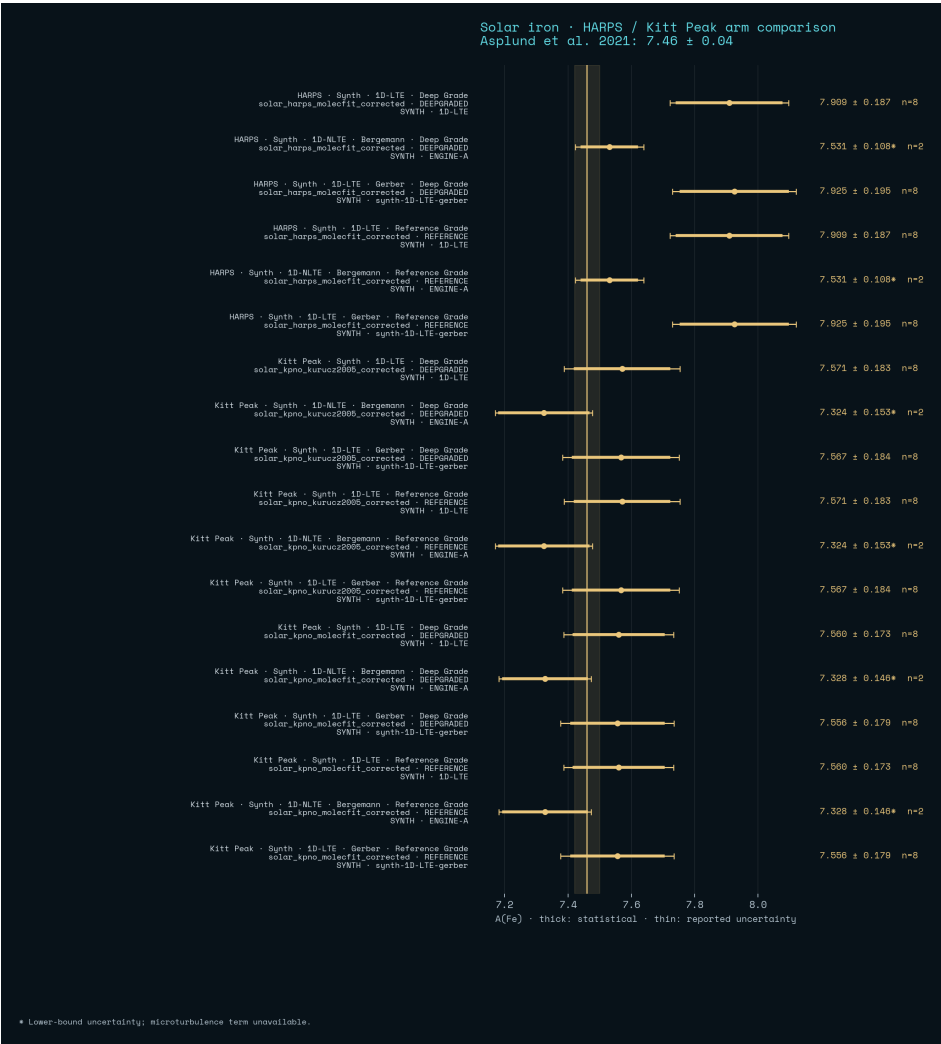
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber ·  $A = 10.0 (+2.5 \text{ dex})$ . RYA-1191 blend audit: 66 catalogued transitions in the 0.62 Å fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6\_atom\_hfs\_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator's fit window; it is not a synthetic spectrum. [Source evidence](#)

### Near-UV molecular opacity



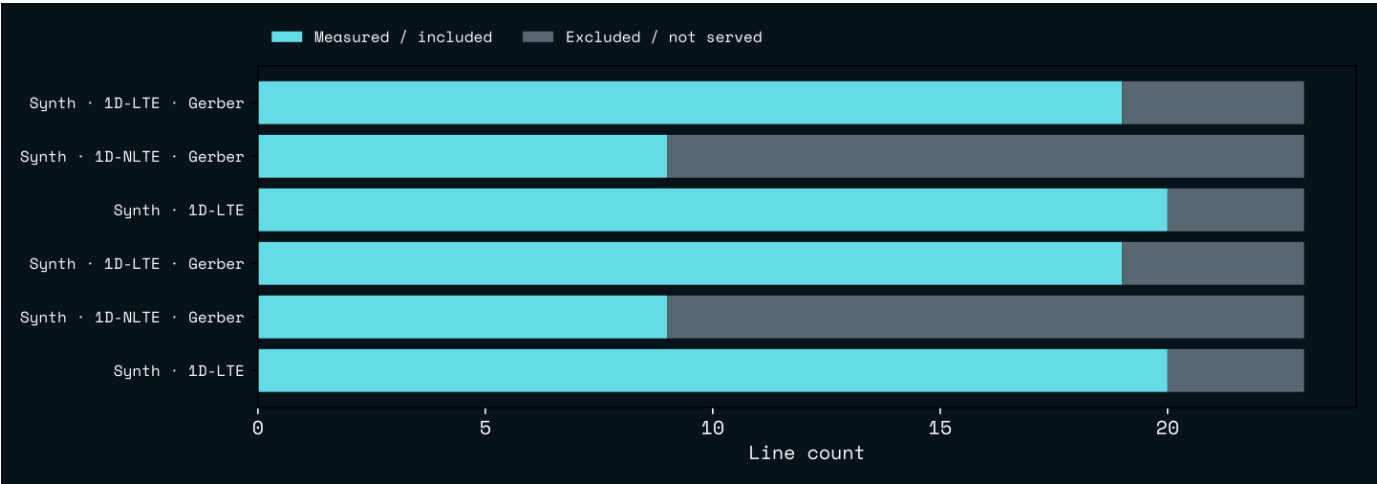
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. [Source evidence](#)

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. [Source evidence](#)

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. [Source evidence](#)

## Download the evidence

6686 graded per-line measurements exported. Evidence for 22 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

[Graded per-line abundances \(CSV\)](#)

[All product fields \(CSV; nested fields preserved as JSON\)](#)

[Per-product line-evidence coverage \(CSV\)](#)

[Source Fe.json](#)

## Reproducibility

Generator scripts/generate\_fe\_publication.py v1.0.0

Science source commit 0e4539da63c8fe63facab98cae5166e98219126b

Fe.json v1.221 · feed timestamp 2026-09-19T05:36:46Z

Generated 2026-09-19T16:52:30+00:00

Feed SHA-256 66421362d81e61e8d3aa6a412e1ae362743c05e41fb8b12a2331347f890395b0

[Refreshed product tracker](#) · [Build manifest](#) · [Fe I appendix](#)

## References

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Lodders, K.; Bergemann, M.; Palme, H. (2025). [Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites](#) (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

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Den Hartog, E. A.; Lawler, J. E.; Sneden, C.; Cowan, J. J.; Brukhovesky, A. (2019). [Atomic Transition Probabilities for UV and Blue Lines of Fe II and Abundance Determinations in the Photospheres of the Sun and Metal-poor Star HD 84937](#) (doi: 10.3847/1538-4365/ab322e). Laboratory transition probabilities used in the resolved Fe II per-line evidence. Coverage and unresolved products are listed in the downloads.